

WEBRUM-05: Error Analysis

Series: WEBRUM — Web Real User Monitoring | **Notebook:** 5 of 10 | **Created:** March 2026 | **Last Updated:** 09/28/2026

Overview

JavaScript errors are among the most impactful issues affecting web application user experience. A single unhandled exception can break page functionality, prevent form submissions, or cause entire features to stop working. Dynatrace RUM captures JavaScript errors, XHR/fetch errors, and custom errors in real-time, enabling you to quantify error impact on real users.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Grail enabled
RUM Enabled	Web applications with error capture active
Permissions	<code>storage:events:read</code>
Previous Notebook	WEBRUM-01: RUM Fundamentals

1. Error Types in RUM

Dynatrace captures several categories of browser-side errors:

Error Type	Description	Example
JavaScript error	Unhandled exception or runtime error	<code>TypeError: Cannot read property 'x' of undefined</code>
XHR error	Failed XHR/fetch request (HTTP 4xx/5xx or network error)	<code>HTTP 500 on POST /api/checkout</code>
Custom error	Errors reported via the RUM API (<code>dtrum.reportError()</code>)	<code>Payment validation failed</code>
CSP violation	Content Security Policy blocking a resource	<code>Refused to load script from 'http://evil.com'</code>
Resource error	Failed resource loading (images, scripts, stylesheets)	<code>Failed to load /assets/app.js</code>

Error Data Model

Field	Description
<code>error.message</code>	Error message text
<code>error.type</code>	Error classification
<code>error.source</code>	Source file and line number

Field	Description
<code>action.name</code>	User action during which the error occurred
<code>sessionId</code>	Session containing the error
<code>application</code>	Application name

```

// Error / navigation vocabulary corrected 08/12/2026 (New RUM):
// filter type == "Error" -&gt; filter characteristics.has_error == true
(11,909 events; identical
// population to isNotNull(error.type), whose
values are request/csp/exception)
// error.message -&gt; error.reason
// user_action.type == "RouteChange" -&gt; "same_view" (the New RUM SPA
route-change value; the
// only other value is "hard_navigation".
"Custom" has NO equivalent.)
// connection.type -&gt; network.protocol.name
// THE EVENT CHARACTERISTIC MATTERS AS MUCH AS THE FIELD: navigation-timing
fields
// (performance.dom_interactive, performance.load_event_end) live on
navigation events
// (characteristics.has_navigation) and are unpopulated on page summaries
// (characteristics.has_page_summary), so a page-summary filter silently
empties them.
// ttfb.* is the opposite – it lives on page summaries. Select with the stable
has_* flags, not
// characteristics.classifier (corrected 09/28/2026): the docs call classifier
"not intended for
// query usage" and Semantic Dictionary 1.349 removes it from the user-event
models.
// Session/performance field vocabulary corrected 08/12/2026 (New RUM).
Classic camelCase RUM
// names are null on New RUM data and fail silently. Verified against 3,261
user.sessions:
// userType -&gt; dt.rum.user_type      userActionCount -&gt;
user_action_count
// totalErrorCount -&gt; error.count     sessionId -&gt; dt.rum.session.id
// hasSessionReplay -&gt; characteristics.has_replay
// browserFamily -&gt; browser.name      osFamily -&gt; os.name
// application -&gt; primary_tags.application
// country/city/continent -&gt; geo.country.name / geo.city.name /
geo.continent.name
// screen.width|height -&gt; browser.window.width|height
// dom.interactive.time -&gt; performance.dom_interactive
// load.event.time -&gt; performance.load_event_end
// server.time -&gt; ttfb.waiting_duration
// TWO TENANT CAVEATS on the validation tenant, both of which leave a CORRECT
query empty:
// * every session is dt.rum.user_type == "synthetic", so a real-user filter
matches nothing –
// the "real_user" literal itself could NOT be confirmed here and is the
documented value form;
// * geo.* is 0-populated, because synthetic traffic carries no geolocation.

```

```
// Field vocabulary corrected 08/12/2026 – this series targets **New RUM**,
// but was written
// against names that are null on New RUM data, so these cells returned
// nothing while erroring
// nowhere. Verified against 5,556,127 user.events records (schema 0.24.0,
// javascript agent):
//   action.type == "Load"           -&gt; characteristics.has_navigation
//                                     == true
//                                     (was characteristics.classifier –
// "not intended for query usage", SD 1.349)
//   action.type                     -&gt; user_action.type
// (hard_navigation | same_view)
//   action.name                     -&gt; page.detected_name
//   web_vitals.largest_contentful_paint -&gt; lcp.start_time      (327,099
// populated)
//   web_vitals.cumulative_layout_shift -&gt; cls.value           (387,254
// populated)
//   app.name                        -&gt; dt.rum.application.id
// UNITS CHANGE WITH THE FIELD. web_vitals.* was a nanosecond DURATION, so `
// 1ms` was correct for
// it; lcp.start_time is a PLAIN NUMBER already in milliseconds, and dividing
// it by 1ms yields
// null. Compare it against the 2500/4000 ms thresholds directly.
// `web_vitals.*` does still exist in the same schema, but carried 16 records
// in 30 days against
// lcp.*'s 327,099 – it is not a different RUM generation, just a rarely-
// populated sibling.
// Recent RUM errors – explore the data structure
fetch user.events, from:-1h
| filter characteristics.has_error == true
| fieldsKeep timestamp, error.reason, error.type, error.source,
page.detected_name, primary_tags.application, dt.rum.session.id
| sort timestamp desc
| limit 20
```

2. Error Volume and Trends

Start by understanding overall error volume and how it changes over time.

```
// Error count by type over last 24 hours
fetch user.events, from:-24h
| filter characteristics.has_error == true
| summarize error_count = count(), by:{error.type}
| sort error_count desc
```

```
// Error trend over 24 hours – hourly bucketed by error type
fetch user.events, from:-24h
| filter characteristics.has_error == true
| makeTimeseries error_count = count(), interval:1h, by:{error.type}
```

```
// Error volume by application – which apps have the most errors?
fetch user.events, from:-24h
| filter characteristics.has_error == true
| summarize error_count = count(),
    unique_errors = countDistinct(error.reason),
    affected_sessions = countDistinct(dt.rum.session.id),
    by:{primary_tags.application}
| sort error_count desc
```

3. Error Grouping and Top Errors

Error messages often contain variable data (stack traces, URLs, IDs). Grouping by error message helps identify the most frequent issues.

```
// Top 15 errors by frequency – the most common error messages
fetch user.events, from:-24h
| filter characteristics.has_error == true
| summarize error_count = count(),
    affected_sessions = countDistinct(dt.rum.session.id),
    by:{error.reason, error.type}
| sort error_count desc
| limit 15
```

```
// Errors by page – which pages generate the most errors?
fetch user.events, from:-24h
| filter characteristics.has_error == true
| filter isNotNull(page.detected_name)
| summarize error_count = count(),
    unique_errors = countDistinct(error.reason),
    by:{page.detected_name}
| sort error_count desc
| limit 10
```

4. Error Impact Analysis

Not all errors are equal. An error affecting 1 session is different from one affecting 10,000. Impact analysis quantifies the blast radius of each error.

```
// Error impact – errors ranked by number of affected sessions
fetch user.events, from:-24h
| filter characteristics.has_error == true
| summarize total_occurrences = count(),
    affected_sessions = countDistinct(dt.rum.session.id),
    by:{error.reason}
| sort affected_sessions desc
| limit 10
```

```
// Error rate per application – percentage of sessions with errors
fetch user.sessions, from:-24h
| filter dt.rum.user_type == "real_user"
| summarize total_sessions = count(),
    error_sessions = countIf(error.count > 0),
    by:{primary_tags.application}
| fieldsAdd error_rate_pct = round(toDouble(error_sessions) /
toDouble(total_sessions) * 100.0, decimals: 2)
| sort error_rate_pct desc
```

Tip: Focus on errors with high session impact rather than high occurrence count. A single user hitting a retry loop can generate thousands of occurrences from one session.

5. XHR and Fetch Errors

XHR/fetch errors indicate backend API failures impacting the frontend. These are particularly critical in SPAs where the UI depends entirely on API responses.

```
// Unit trap (08/12/2026): New RUM timing fields such as lcp.start_time and
tftb.waiting_duration
// are PLAIN NUMBERS already in milliseconds, not durations. `field / 1ms`
yields null on them –
// silently, so a chart of nulls looks like "no data". Compare and aggregate
them directly.
// XHR errors by action – identify failing API calls
fetch user.events, from:-24h
| filter characteristics.has_error == true
| filter error.type == "request"
| summarize error_count = count(),
    affected_sessions = countDistinct(dt.rum.session.id),
    by:{error.reason, page.detected_name}
| sort error_count desc
| limit 15
```

```
// XHR error trend – are backend errors increasing?
fetch user.events, from:-24h
| filter characteristics.has_error == true
| filter error.type == "request"
| makeTimeseries error_count = count(), interval:1h
```

6. Rage Click Detection

A **rage click** occurs when a user rapidly clicks the same element multiple times — a strong signal of frustration. This usually indicates:

- A button that appears clickable but is not responding
- A form submission that is silently failing
- A UI element that is loading too slowly
- A dead link or broken navigation element

Dynatrace captures rage clicks as user action properties. They can also be detected through rapid action sequences:

```
// Sessions with rage clicks – identify frustrated users
fetch user.events, from:-24h
| filter type == "RageClick"
| summarize rage_count = count(),
  by:{page.detected_name, primary_tags.application}
| sort rage_count desc
| limit 10
```

```
// Rage click trend over 7 days – is frustration increasing?
fetch user.events, from:-7d
| filter type == "RageClick"
| makeTimeseries rage_count = count(), interval:1d
```

7. Error-to-Session Correlation

Understanding how errors correlate with session outcomes (bounce, low engagement, failed conversion) helps prioritize fixes.

```
// Compare sessions with and without errors – engagement impact
fetch user.sessions, from:-24h
| filter dt.rum.user_type == "real_user"
| fieldsAdd has_errors = if(error.count > 0, "With Errors", else: "No Errors")
| summarize session_count = count(),
  avg_actions = avg(user_action_count),
  avg_duration_sec = avg(duration / 1s),
  bounce_rate = countIf(user_action_count == 1),
  by:{has_errors}
| fieldsAdd bounce_pct = round(toDouble(bounce_rate) / toDouble(session_count)
  * 100.0, decimals: 1)
```

```
// Errors by browser – are certain browsers more error-prone?
fetch user.sessions, from:-24h
| filter dt.rum.user_type == "real_user"
| filter isNotNull(browser.name)
| summarize total = count(),
    with_errors = countIf(error.count > 0),
    by:{browser.name}
| fieldsAdd error_rate = round(toDouble(with_errors) / toDouble(total) *
100.0, decimals: 1)
| filter total > 10
| sort error_rate desc
| limit 10
```

8. Summary and Next Steps

In this notebook, we covered:

- **Error types** — JavaScript errors, XHR errors, custom errors, resource errors
- **Error volume and trends** — Tracking error rates over time
- **Error grouping** — Identifying the most frequent error messages
- **Impact analysis** — Measuring how many sessions each error affects
- **XHR/fetch errors** — Backend API failures visible from the frontend
- **Rage clicks** — Detecting user frustration through rapid repeated clicks
- **Error-session correlation** — How errors impact engagement and bounce rates

Next Steps

- **WEBRUM-06: Performance Analysis** — Page load waterfall and performance bottleneck identification
- **WEBRUM-07: Session Replay** — Visually investigate error-impacted sessions

References

- [Configure error detection for web applications \(DT docs\)](#)
- [User actions in RUM Classic \(DT docs\)](#)
- [User events — semantic dictionary \(DT docs\)](#) — *"Used for internal optimization when storing the data and not intended for query usage."*
- [Semantic Dictionary changelog 1.349 \(DT docs\)](#)

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